

## Advanced (Habanero)

**Q1.** What is the vertex of the graph of  $y = |x - 2|$ ?

- (A) (0, 2)
- (B) (2, 0)
- (C) (2, 2)
- (D) (0, 0)
- (E) (1, 1)

**Q2.** What is the axis of symmetry of the graph of  $y = |x + 1|$ ?

- (A)  $x = -1$
- (B)  $x = 1$
- (C)  $x = 0$
- (D)  $y = -1$
- (E)  $y = 1$

**Q3.** The graph of  $y = |x|$  opens:

- (A) Up
- (B) Down
- (C) Left
- (D) Right
- (E) None

**Q4.** What is the equation of the graph of  $y = |x - 3| + 2$  shifted 1 unit to the left?

- (A)  $y = |x - 2| + 2$
- (B)  $y = |x - 4| + 2$
- (C)  $y = |x + 1| + 2$
- (D)  $y = |x - 3| + 1$
- (E)  $y = |x + 3| - 2$

**Q5.** Which of the following is the graph of  $y = 2|x|$ ?

- (A) A narrow V-shape
- (B) A wide V-shape
- (C) A straight line
- (D) A parabola
- (E) A circle

**Q6.** What is the effect of adding 3 to the function  $y = |x|$ ?

- (A) Shifts the graph up 3 units
- (B) Shifts the graph down 3 units
- (C) Shifts the graph left 3 units
- (D) Shifts the graph right 3 units
- (E) Stretches the graph vertically

**Q7.** The equation  $y = |x - 2| - 1$  represents a graph that is:

- (A) Shifted 2 units to the right and 1 unit down
- (B) Shifted 2 units to the left and 1 unit up
- (C) Shifted 2 units to the right and 1 unit up
- (D) Shifted 2 units to the left and 1 unit down
- (E) Shifted 1 unit to the right and 2 units down

**Q8.** What is the effect of multiplying the function  $y = |x|$  by  $-1$ ?

- (A) Flips the graph vertically
- (B) Flips the graph horizontally
- (C) Stretches the graph vertically
- (D) Compresses the graph vertically
- (E) Shifts the graph up 1 unit

**Open Ended Questions (FRQ)****Q9.** Graph the function  $y = |x + 2| - 3$ . Identify the vertex, axis of symmetry, and direction of opening. Provide a brief explanation.**Q10.** Compare and contrast the graphs of  $y = |x|$  and  $y = |x| + 2$ . Describe any shifts, stretches, or flips. Provide a sketch of both graphs.**Chef's Tips**

- Provide clear and concise instructions
- Emphasize the importance of labeling axes and vertices
- Encourage students to check their work